and discharging of lithium-ion batteries include:

Charge current. Charge current must be limited for lithium-ion batteries with a typical maximum value of 0.8C, but lower values are usually set to give some margin.

Charge temperature. Lithium-ion battery charge temperature should be monitored, and cell or battery must not be charged when temperature is lower than 0°C or greater than 45°C.

Discharge current. Discharge current protection is required to prevent damage or explosion as a result of short-circuits.

Over-voltage. Charge over-voltage protection is required to prevent voltage that is too high to be applied across battery terminals.

Over-charge protection. Overcharge protection circuitry is required to stop the lithium-ion charging process when voltage per cell rises above 4.3 volts.

Reverse polarity protection. Lithium-ion battery reverse polarity protection is needed to make sure the battery is not charged in the wrong direction as this could lead to serious damage or even explosion.

Lithium-ion over-discharge. Overdischarge protection is required to prevent battery voltage falling below about 2.3 volts or dependent on manufacturing of the cell.

Over-temperature. Over temperature protection is often incorporated to prevent the battery from operating if temperature rises too high, and temperatures above 100°C can cause irreparable damage to the battery.

Battery protection challenges

If the battery gets hot enough to ignite the electrolyte, a fire is usually caused by an internal short in the battery. Lithium-ion cells contain a separator sheet that keeps positive and negative electrodes apart. If that sheet gets punctured and electrodes touch, the battery heats up very quickly. Heat produced can cause the battery to vent the organic electrolyte or light it. In a separator failure, some kind of short happens inside the lithium-ion battery. Once that happens inside one of the cells, heat of the fire cascades to other cells, and the whole pack goes up in flames.

Lithium battery packs are particularly sensitive to faults caused by external shorts, runaway charging conditions and abusive overcharging, which can cause potentially damaging over-current and over-temperature conditions. Hence, there is a need to enforce safety regulations and established test requirements for lithium-ion and lithium-polymer packs to demonstrate their resilience to both short-circuit and overcharge events.

Protection circuits built into the battery pack limit the peak voltage of each cell during charging as well as prevent cell voltage from dropping too low on discharging. In addition, cell temperature is also monitored with the help of protection circuits acting as temperature sensors to prevent temperature extremes as maximum charge and discharge current on most pacts is limited to between 1C and 2C. Using such precautions, the possibility of metallic-lithium plating due to overcharging is virtually eliminated. A few battery protection methods are summarised below.

- Temperature sensors to monitor battery temperature.
- Voltage converter and regulator circuit to maintain safe levels of voltage and current.
- Shielded notebook connector that lets power and information flow in and out of battery pack.
- Voltage tap to monitor energy capacity of individual cells in battery pack.
- Battery charge state monitor to handle the whole charging process to make sure battery charges as quickly and fully as possible.
- Battery storage in a cool place that slows the aging process of lithium-ion (and other chemistries).

Future scope

Twenty-five years ago, the lithium-ion battery made its debut into the marketplace as a result of innovative work by Asahi Kasei, and development and marketing by Sony Corp. Since the realisation of lithium-ion batteries, there has been tremendous development in terms of their capacity, energy, power and cost reduction.

Lithium-ion batteries have become indispensable today when we talk about the life quality of people in modern society. It can easily be called a dominant technology that is widely used in portable electronic devices. Lithium-ion batteries are also a preferred option today in the emerging sector of EVs. The technology is still on the breakthrough for its use in the field of power supply systems like wind power and photovoltaics. Thus, the technology offers a massive global potential in terms of reduction in carbon emissions and energy sustainability. Although these batteries show a lot of promise to rid the world of several challenges in the future, there are still a few shortcomings of this battery that need to be addressed.

Safety of lithium-ion batteries remains an important concern for the industry. However, new developments in separator technology have improved the outlook towards safer batteries. Similarly, with recent progress in new materials and successful implementation of new battery concepts in active materials, inert materials and cell designs, the lithium-ion battery will continue to improve in all its properties.

Researchers and manufacturers are constantly improving on lithium-ion batteries, introducing new and enhanced chemical combinations every six months or so. With such rapid progress, it will be possible to develop new lithium-ion batteries free from existing disadvantages.

Finally, several investigators are studying the possibility of 3D architecture of lithium-ion battery structures including porous or expanded metal collectors. This would help to increase battery density and spatial utilisation if production-friendly concepts are developed. The result of these advances, if realised, could yield lithium-ion battery with specific energies of 400Wh/kg with at least moderate power density. This would represent an increase of about sixty per cent compared to the best of today's 18650 cells with 3400mAh capacity, and could come about within the next several years. Hence, the future for lithium-ion battery continues to be bright, especially if manufacturers are careful to maintain safe practices in manufacturing processes and new designs. EFY